

Activity-I

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① Medical Image Enhancement:

1. which gray-level transformation could you recommend?

1. log transformation.

2. why is it suitable?

A - log transformation enhances the darker regions of the x-ray image about without greatly affecting the lighter bone region.

3. what effect will it have on the image?

A - The dark tissues become brighter and cleared improving visibility while the bones remain visible with better overall contrast.

② Night-time Surveillance:

1. which transformation would improve the visibility.

= log transformation.

2. why is it more suitable than negative transformation.

- log transformation increases the brightness of dark pixels, making objects easier to see

3) Fingerprint recognition:-

1. which gray-level transformation could you apply?

A. Contrast stretching

2. what would happen if a log transformation were applied instead?

=> log transformation would brighten the darker regions but would not improve the ridge-to-valley contrast as effectively as contrast stretching.